CCGM1A1 Characteristics

VDDIO = 1.1~2.7V (I/O supply voltage (GPIO banks and VDD_CLK))
VDDcore = 0.85~1.15V (Core supply voltage (VDD, VDD_PLL, VDD_SER and VDD_SER_PLL))
VDDSER = 0.95~1.15V (SerDes supply voltage (VDD_SER))
VDDSER_PLL = 0.95~1.15V (SerDes PLL supply voltage (VDD_SER_PLL))

Unequal Pins A1/A2/A4!

Ball U16 is SER_TX_P1 in A2/A4!
Ball V16 is SER_TX_N1 in A2/A4!
Ball U14 is SER_RX_P1 in A2/A4!
Ball V14 is SER_RX_N1 in A2/A4!
Ball N16 is VDD_SER in A4!

Note!: Take in mind that SerDes is not working when VDD_CORE = 0.9V (Low power mode), because VDDSER and VDDSER_PLL have got minimum working voltage: 0.95V!

Power-On Sequence:
VDD_CORE -> +2V5 -> +3V3 -> +1V8 -> +1V2

VDD_CORE_SET1	VDD_CORE (Mode)	fDCO (MHz)	fPLL (MHz)
Both Opened	0.90V (Low power mode)	500~1000	up to 250
1-2 Closed	1.00V (Economy mode)	1000~2000	up to 312.5
2-3 Closed	1.10V (Speed mode)	1250~2500	up to 416.75

Notes:
- Supports 3 classes: A,B,C
- Different Supply Voltages: 5V, 3.3V, 1.8V
- three AS – Application Specific pins
- does not support +21V DC [Programming]

If used as Master:
- Place all components
- 5V direct pass though of the USB Voltage
- 3.3V, 1.8V 500mA current limited

Vout	EA_B7	EA_A7
5V	LOW	–
3V3	HIGH	Z
1V8	HIGH	LOW

RP2040 Characteristics

IOVDD = 1.8~3.3V
DVDD = 1.1V
VREG_VIN = 1.8~3.3V
USB_VDD = 3.3V
ADC_AVDD = 3.3V

Notes:
1. RP2040 supports 5MHz to 15MHz crystals(XOSC) to produce an accurate reference clock(clk_ref). The reference clock is distributed to the PLLs, which can be used to multiply the XOSC frequency to provide accurate high speed clocks. For example, they can generate a 48MHz clock which meets the frequency accuracy requirement of the USB interface and a 133MHz maximum speed system clock. clk_ref = (6-12)MHz(Nominal) – Reference clock that is always running unless in DORMANT mode. Runs from Ring Oscillator (ROSC) at power-up, but can be switched to Crystal Oscillator (XOSC) for more accuracy.
2. There are two PLLs in RP2040. They are: pll_sys – Used to generate up to a 133MHz system clock(clk_sys), pll_usb – Used to generate a 48MHz USB reference clock(clk_usb).
3. System clock clk_sys = 125MHz(Nominal), Runs from clk_ref at power-up but is typically switched to a PLL.
4. Peripheral clock clk_peri = (12-125)MHz(Nominal). Typically runs from clk_sys but allows peripherals to run at a consistent speed if clk_sys is changed by software.
5. The maximum SPI clock frequency is half the system clock frequency(clk_sys). For example, at the maximum SSPCLK (clk_peri) frequency on RP2040 of 133MHz, the maximum peak bit rate in master mode is 62.5Mbps. In slave mode, the same maximum SSPCLK frequency of 133MHz can achieve a peak bit rate of 133 / 12 = ~11.083Mbps.
6. CLOCK GPINx – General purpose clock inputs. Can be routed to a number of internal clock domains on RP2040, e.g. to provide a 1Hz clock for the RTC, or can be connected to an internal frequency counter. GPIN0, GPIN1 and XIN inputs are limited to 50MHz.
7. CLOCK GPOUTx – General purpose clock outputs. Can drive a number of internal clocks (including PLL outputs) onto GPIOs, with optional integer divide. clk_gpout[0..3] – Clock output to GPIO. Can be used to clock external devices or debug on chip clocks with a logic analyser or oscilloscope.

https://www.olimex.com/
based on GateMateA1-EVB

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Title: MillerV1

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